Train a Classifier

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Recall and choose your starting point

> Record your starting explanation for Train a Classifier before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Nearest Neighbors

Without opening the old lesson, explain one key idea from Nearest Neighbors. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Baselines First

Without opening the old lesson, explain one key idea from Baselines First. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Sampling and Intended Use

Without opening the old lesson, explain one key idea from Sampling and Intended Use. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 26

Practise with fading help

> Complete the supported Train a Classifier trace and record the first step you can explain without help.

Use W2: Trace and score the saved run for A reproducible classifier **pipeline**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 26

Individual reasoning check

> Check the proposed Train a Classifier explanation against the supplied facts before discussing it.

Use the worked case for A reproducible classifier **pipeline**. Proposed explanation: “A **pipeline** prevents every leak automatically”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 26

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Build and trace a fresh fitted state

Use P (0,60) blocked, Q (20,20) clear, R (20,60) clear as new **training** rows. Fit the same **pipeline**. **Predict** validation queries (2,28),(18,56),(4,12) with references blocked,clear,blocked.

Give fitted minima and ranges, transformed queries, selected IDs, **prediction**s, accuracy and **training**-majority baseline.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 26

Your learning target

> Connect the Train a Classifier target to a trace and a justified conclusion.

Training: spots 0:H,2:H,4:S,6:S. Compare rules S if spots>=1,>=3,>=5. Choose by **training** accuracy, then **predict** held-out 1:H and 5:S without revising. Report both stages.

WORKBOOK · 6 OF 26

Solve the given case

> Solve the supplied Train a Classifier case and show the deciding evidence.

Training is (0,H),(1,H),(4,S),(5,S). **Candidate rule predict**s S for spots>=3. Held-out cases are (2,H),(3,S).

Apply the rule and distinguish **fitting** from inference.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

A classifier fits on rows 1–80 and is evaluated on those same rows. Its accuracy is 100%. What additional **evidence** is needed before claiming generalization?

My independent answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 26

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 26

Project checkpoint

> Justify what Train a Classifier adds to Responsible Plant Classifier and keep the evidence.

Train Plant Classifier v1 without opening the final test labels during model choice; save the experiment record.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 26

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 26

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Training: spots 0:H,2:H,4:S,6:S. Compare rules S if spots>=1,>=3,>=5. Choose by **training** accuracy, then **predict** held-out 1:H and 5:S without revising. Report both stages.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

W1 Separate fit predict and evaluate

> Separate fit predict and evaluate. Apply the supplied Train a Classifier facts and keep your reasoning.

List what fit learns or stores, what **predict** takes as **input**, and what evaluation compares. State where the validation reference labels belong.

My reasoning and evidence

W2 Trace and score the saved run

> Trace and score the saved run. Apply the supplied Train a Classifier facts and keep your reasoning.

Give the IDs and labels printed for V1,V2,V3. Use references clear,blocked,clear to calculate accuracy and the **training**-majority baseline. Which error should repeat in an unchanged rerun?

My reasoning and evidence

W3 Repair an incomplete run record

> Repair an incomplete run record. Apply the supplied Train a Classifier facts and keep your reasoning.

A classmate saves only k=1 and a score of 2/3. Name the information needed to **reproduce** this run. Explain why the same score alone is insufficient.

My reasoning and evidence

WORKBOOK · 15 OF 26

W4 Build and trace a fresh fitted state

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Build and trace a fresh fitted state. Apply the supplied Train a Classifier facts and keep your reasoning.

Rules supplied for this independent case

Pipeline rules: fit each feature's minimum and range (maximum minus minimum) using training rows only. Scale every training and query component as (value – fitted minimum) / fitted range, without clipping. For k=1, sum absolute component differences and break distance ties by alphabetical ID. The baseline is the training-majority label. Do not refit on validation queries.

Use P (0,60) blocked, Q (20,20) clear, R (20,60) clear as new **training** rows. Fit the same **pipeline**. **Predict** validation queries (2,28),(18,56),(4,12) with references blocked,clear,blocked.

Give fitted minima and ranges, transformed queries, selected IDs, **prediction**s, accuracy and **training**-majority baseline.

My reasoning and evidence

W5 Diagnose pipeline misuse

Diagnose pipeline misuse. Apply the supplied Train a Classifier facts and keep your reasoning.

Explain and repair three mistakes: **predict** receives an already scaled query; fit receives **training** plus validation rows; a query uses brightness before distance. Does repeating each mistaken run make it valid?

My reasoning and evidence

Searching candidate rules — W1 State the search before running it

> Use the supplied Train a Classifier task to record your reasoning, evidence and next step.

Write the rule direction, equality behavior, candidate cutoffs and tie policy. Identify what humans supplied and what **fitting** selects.

My reasoning and evidence

Searching candidate rules — W2 Trace the training mistakes

> Use the supplied Train a Classifier task to record your reasoning, evidence and next step.

List errors for cutoffs 3, 5 and 7 on the worked **training** rows. Identify the mistaken row for each imperfect candidate. Which cutoff is stored?

My reasoning and evidence

WORKBOOK · 19 OF 26

Searching candidate rules — W3 Audit the boundary and comparison

> Use the supplied Train a Classifier task to record your reasoning, evidence and next step.

What happens at distance 5 if **prediction** uses < instead of <= with cutoff 5? Explain why the search uses a strict < for error counts even though **prediction** uses <= for distance.

My reasoning and evidence

WORKBOOK · 20 OF 26

Searching candidate rules — W4 Search and evaluate independently

> Use the supplied Train a Classifier task to record your reasoning, evidence and next step.

New **training** rows are (1,blocked), (3,clear), (5,blocked), (7,clear). Try cutoffs 2,4,6, using the same direction and smaller-cutoff tie policy. Give every **training error** count and select a cutoff.

Validation rows are (2,blocked), (4,blocked), (6,clear), (8,clear). List the selected rule's **prediction**s and score. Compare with the **training**-majority baseline using clear on ties.

My reasoning and evidence

WORKBOOK · 21 OF 26

Searching candidate rules — W5 Limit the claim

> Use the supplied Train a Classifier task to record your reasoning, evidence and next step.

The worked rule fits **training** perfectly but ties the baseline on validation accuracy. Explain why neither result proves it is the best possible model. Name two details to keep in the experiment record.

My reasoning and evidence

WORKBOOK · 22 OF 26

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 23 OF 26

Pictorial case practice: Train a Classifier

> Separate a reproducible fitted pipeline from merely displaying predictions. Explain the labelled step before checking the changed condition.

Separate a reproducible fitted **pipeline** from merely displaying **prediction**s.

Toy **training** lengths are [2, 6] cm with labels [healthy, spotted]. **Pipeline** converts cm to mm, then uses k=1 absolute distance. Query is 5 cm.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/13

Train a Classifier

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Fit transform

The declared conversion produces training lengths 20 and 60 mm.

2. Transform query

5 cm becomes 50 mm using the same conversion.

3. Predict neighbour

Distances 30 and 10 mm select the spotted training record.

CHANGE ONE THING

Transform training but leave query numerically at 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which **pipeline** record would help another team **reproduce** the result? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

WORKBOOK · 24 OF 26

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 25 OF 26

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Separate a reproducible fitted **pipeline** from merely displaying **prediction**s.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Distances 30 and 10 mm select the spotted **training** record.

1. **Predict** neighbour

2. Fit transform

3. Transform query

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: **Predict** neighbour Distances 30 and 10 mm select the spotted **training** record. Read the picture in order. Distances 30 and 10 mm select the spotted **training** record.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Transform **training** but leave query numerically at 5. Which

conclusion is supported by the supplied facts?

1. Distances become 15 and 55, incorrectly selecting healthy.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Distances become 15 and 55, incorrectly selecting healthy. The query is now compared in inconsistent units; the changed result is a **pipeline** defect.

Repair a missing building block

Review Nearest Neighbors (ai-4/12). Goal: Compute absolute distances and rank **training** neighbours. Short practice: Why is the k=1 outcome at length5 not uniquely determined without a tie rule?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Baselines First (ai-4/10). Goal: Fit a majority label baseline using **training** labels only. Short practice: Can the candidate's **training** accuracy replace this validation comparison?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Sampling and Intended Use (ai-4/7). Goal: Inspect row meaning missing values and repeated records. Short practice: Which collection would address the intended-use gap most directly?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. **Predict** first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

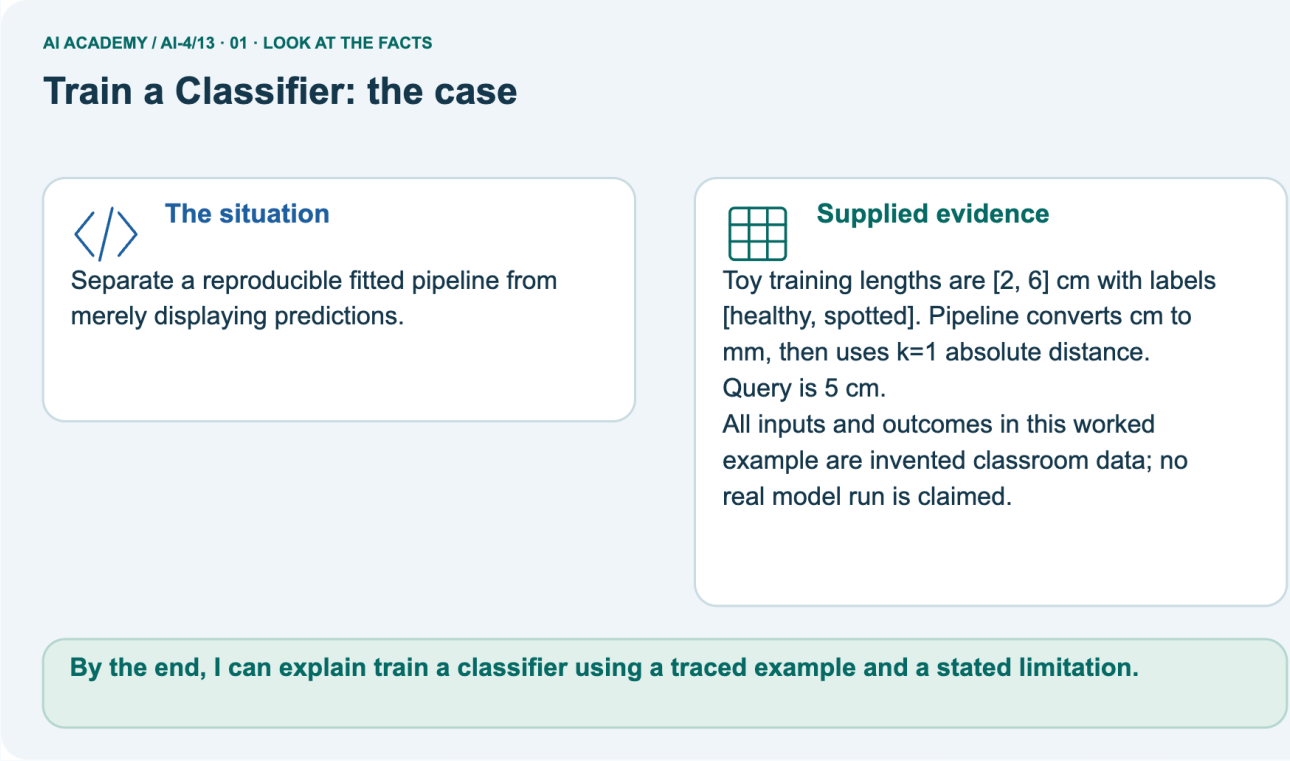
What I repaired, the evidence I kept and the next test

WORKBOOK · 26 OF 26

Visual concept journey

> Explain each Train a Classifier picture from its labels, then record a relevant limit.

01. Train a Classifier: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

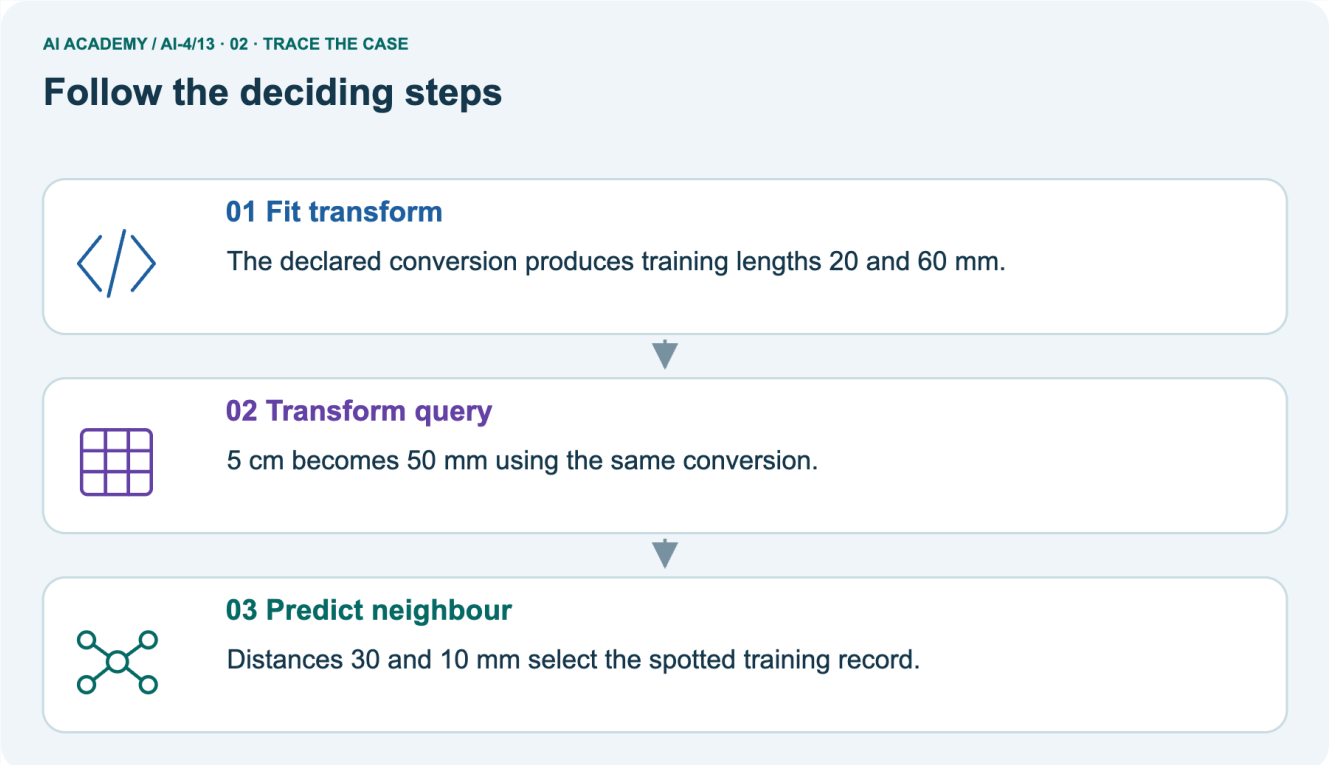
Separate a reproducible fitted **pipeline** from merely displaying **prediction**s. Supplied facts: Toy **training** lengths are [2, 6] cm with labels [healthy, spotted]. **Pipeline** converts cm to mm, then uses k=1 absolute distance.

Query is 5 cm. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



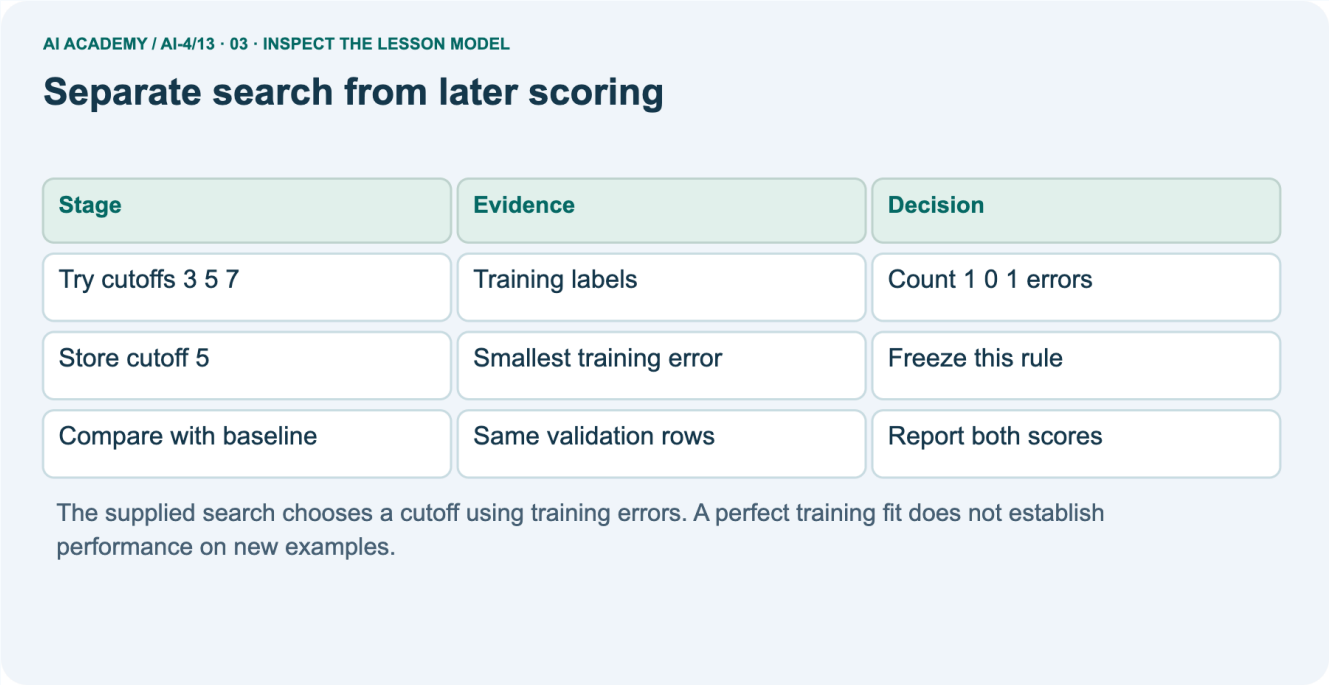
Follow the labelled picture with the matching explanation. Open the vector diagram

Fit transform: The declared conversion produces **training** lengths 20 and 60 mm. → Transform query: 5 cm becomes 50 mm using the same conversion. → **Predict** neighbour: Distances 30 and 10 mm select the spotted **training** record.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Separate search from later scoring



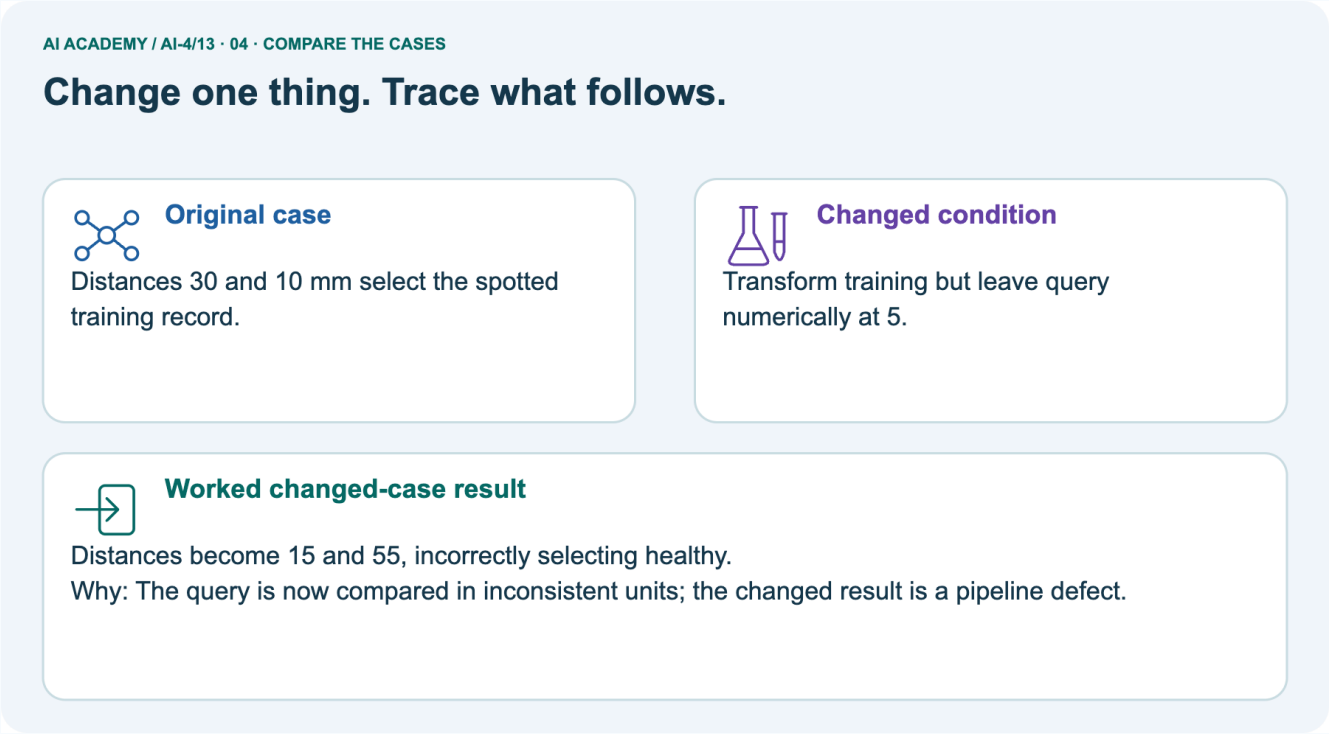
Follow the labelled picture with the matching explanation. Open the vector diagram

Separate search from later scoring. Try cutoffs 3 5 7 | **Training** labels | Count 1 0 1 errors. Store cutoff 5 | Smallest **training error** | Freeze this rule. Compare with baseline | Same validation rows | Report both scores

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

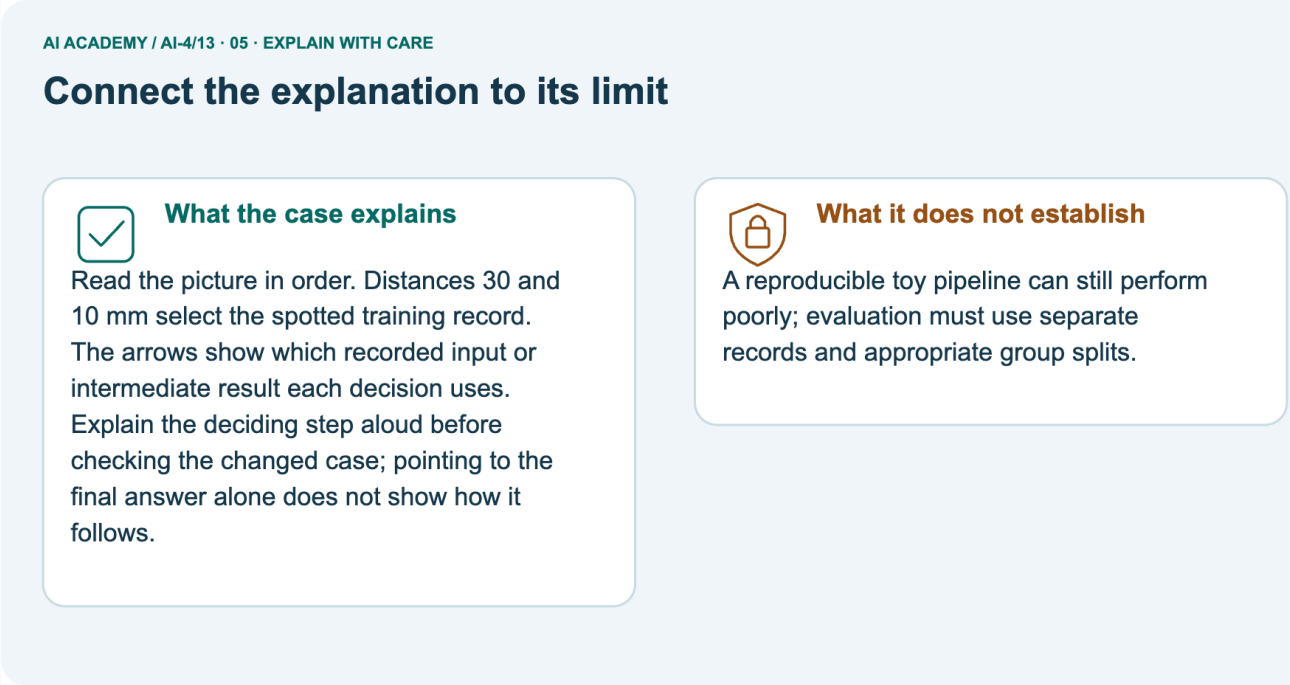
Original case: Distances 30 and 10 mm select the spotted **training** record. Changed condition: Transform **training** but leave query numerically at 5. Expected result: Distances become 15 and 55, incorrectly selecting healthy.

Reason: The query is now compared in inconsistent units; the changed result is a **pipeline** defect.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Distances 30 and 10 mm select the spotted **training** record. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A reproducible toy **pipeline** can still perform poorly; evaluation must use separate records and appropriate group splits.

> Explain the case, then point to a limit on the claim.

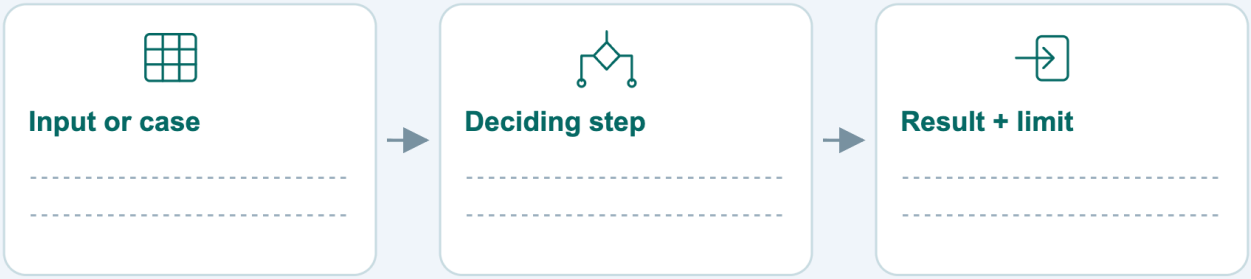
My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project

AI ACADEMY / AI-4/13 · 06 · DRAW AND EXPLAIN

Make the idea visible in your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Which pipeline record would help another team reproduce the result? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which **pipeline** record would help another team **reproduce** the result? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.